

1. The 3×3 reduced row echelon matrix derives from the original matrix A 's pivots. Given that this is a square matrix, the R matrix contains only column such that every other element is zero. The $\vec{x}$ vector contains the solution to $A\vec{x} = \vec{b}$.

* Question 1.c. was asking for numerical values. Points will not be subtracted as they are found here (circled)

-5
b. $A\vec{x} = \vec{b}$
 $\Rightarrow \vec{x} = A^{-1}\vec{b}$

$\Rightarrow d = \begin{bmatrix} 4 \\ 0 \\ 0 \end{bmatrix}$

$\Rightarrow \begin{bmatrix} 4 \\ 0 \\ 20 \end{bmatrix} \Rightarrow \begin{bmatrix} 4 \\ 12 \\ 20 \end{bmatrix}$

$b = \begin{bmatrix} 4 \\ 12 \\ 20 \end{bmatrix}$

$\begin{bmatrix} 0 & 1 & 2 & 2 \\ 0 & 3 & 8 & 7 \\ 0 & 0 & 4 & 2 \end{bmatrix}$

-9
a. $\Rightarrow \begin{bmatrix} 0 & 1 & 2 & 2 \\ 0 & 3 & 8 & 7 \\ 0 & 0 & 4 & 2 \end{bmatrix} = \begin{bmatrix} 0 & 1 & 2 & 2 \\ 0 & 0 & 2 & 1 \\ 0 & 0 & 0 & 0 \end{bmatrix} \Rightarrow$

b. the column space consists of two independent columns such that their combinations fill the rest of the columns.

As such, $C(A) = \left\{ \begin{bmatrix} 1 \\ 3 \\ 0 \end{bmatrix}, \begin{bmatrix} 2 \\ 8 \\ 4 \end{bmatrix}, \begin{bmatrix} 2 \\ 7 \\ 2 \end{bmatrix} \right\}$

$\begin{bmatrix} 0 \\ 0 \end{bmatrix}$ can be formed from the column space above.

Using the nullspace solutions:

$c_3x + c_1x + 0 = 0$

$c_3x + c_2x = 0$

$2c_3 + c_1 = 0$

$5c_3 + c_2 = 0$

* there is only one pivot.

$\Rightarrow A = \begin{bmatrix} 1 & -2 & -5 \\ 3 & -6 & -15 \\ 5 & -10 & -25 \end{bmatrix}$

$\Rightarrow R = \begin{bmatrix} 1 & -2 & -5 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$

E_{31}, E_{21}

Since only column 1 contains a "free" variable, the only special solution is:

$\begin{bmatrix} c \\ 0 \\ 0 \end{bmatrix}$ where c is any real number.

As such, the nullspace are all combinations of the first column plus zero.

$\begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$ (only for 2, 2, 2)

$B = \begin{bmatrix} A & A \\ A & A \end{bmatrix}$

$\Rightarrow \begin{bmatrix} A & A \\ 0 & 0 \end{bmatrix} \Rightarrow \begin{bmatrix} 0 & 1 & 2 & 2 & 0 & 3 & 8 & 7 \\ 0 & 3 & 8 & 7 & 0 & 0 & 4 & 2 \\ 0 & 0 & 4 & 2 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$

Asking for $\vec{y}$ ref! Double! A what?

4. $A = \begin{bmatrix} 2 & 5 \\ 6 & 4 \end{bmatrix}$ Since we know the solutions to $A\vec{x} = \vec{b}$,

these form the column space by which any combination (combination) give the remaining solutions. This is because they form their own space by which solutions $\vec{b}$ are constructed.

b. Yes!

$3a_1 + 2a_2$
 $\Rightarrow 3 \begin{bmatrix} 2 \\ 6 \end{bmatrix} + 2 \begin{bmatrix} 5 \\ 4 \end{bmatrix} = \begin{bmatrix} 18 \\ 26 \end{bmatrix}$

3. A is a 3×5 matrix with $r=3$.

a. Then the equation $A\vec{x} = \vec{b}$ has many solutions.

always

worked differently but works...

b. the column space of A consists of three independent columns that form the remaining columns. the nullspace consists of the zero vector plus the amount of special solutions - which would be 2 given the number of free variables.